LESSON PLAN

QA/QC AND INDIGENISATION (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	5	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Constant Speed Unit (CSU)
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Constant Speed Unit (CSU).
5. Specific Objectives	11.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of CSU cutaway model diagrams; Safety: Do not touch moving parts; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propeller?
(b) How does a fixed-pitch propeller differ from a variable-pitch propeller?
(c) Why is a Constant Speed Unit required for efficient flight?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of Propellers • Fixed Pitch Propeller • Variable Pitch Propeller • Hydrodynamic Principles • Blade Angle Definitions	L & Ds	PPT, IB	15 min
(b)	• CSU Internal Mechanism • Governor Operation • Oil Pressure Control • Flyweight Centrifugal Force • Pitch Change Sequence	L & Ds	PPT, PDF	20 min
(c)	• High Pitch vs Low Pitch • RPM Regulation • Throttle vs Mixture • CSU Failure Modes • System Limitations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the governor maintains constant RPM during flight.
(b) Describe the mechanism that changes propeller pitch in response to oil pressure.
(c) Identify two common failure modes of a Constant Speed Unit.

12. Home Assignment and Follow Up	Prepare a brief note on the effect of propeller rotation on aircraft and methods used to counteract it, as covered in LP 05.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Effect of propeller rotation on aircraft and methods used to counteract it.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min